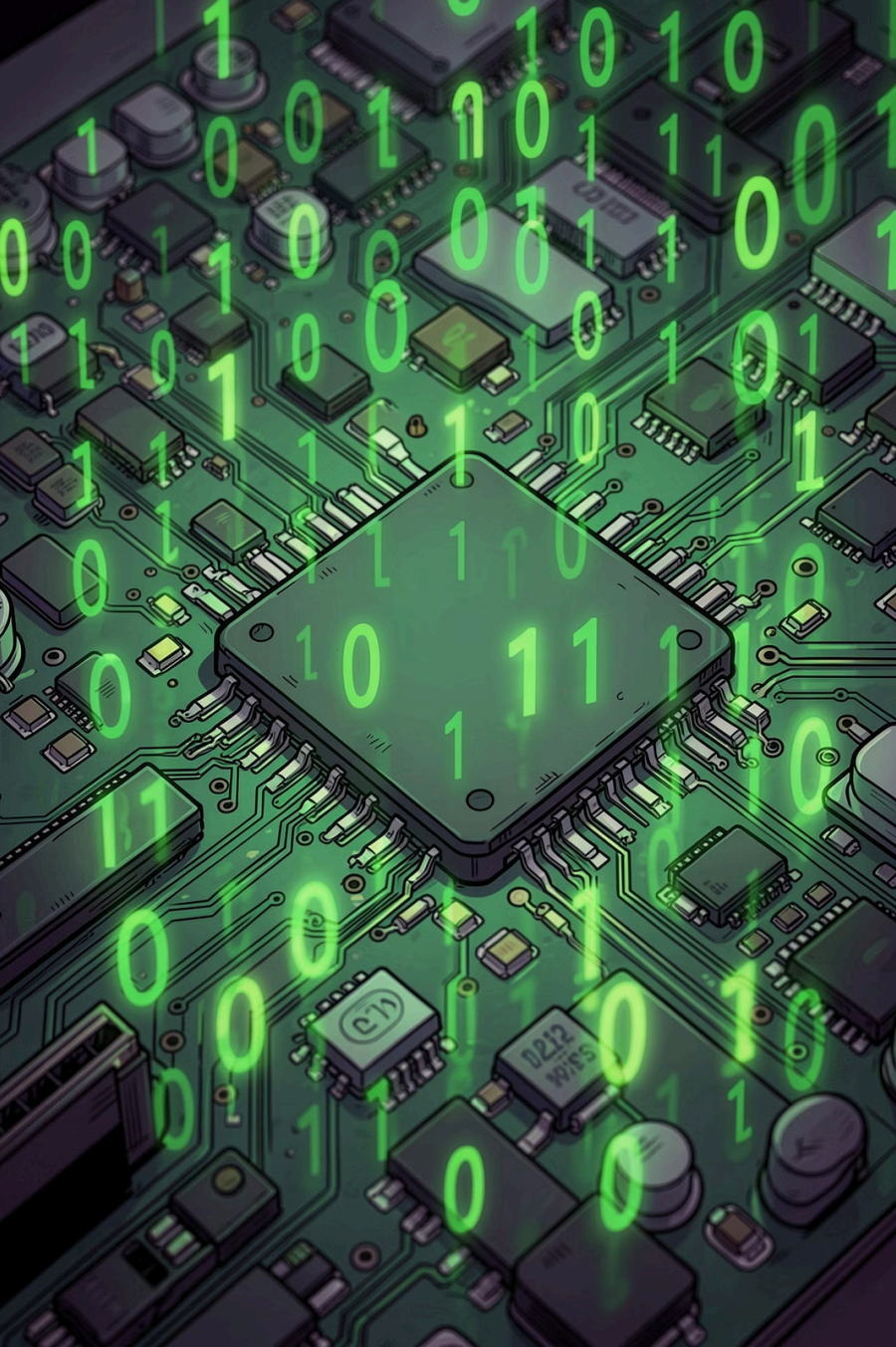




Comparison Between Hamming Code and Polar Code

A concise, 5-minute technical overview comparing two foundational error-correcting codes used in digital communications: Hamming (simple linear block codes) and Polar (capacity-achieving modern codes).

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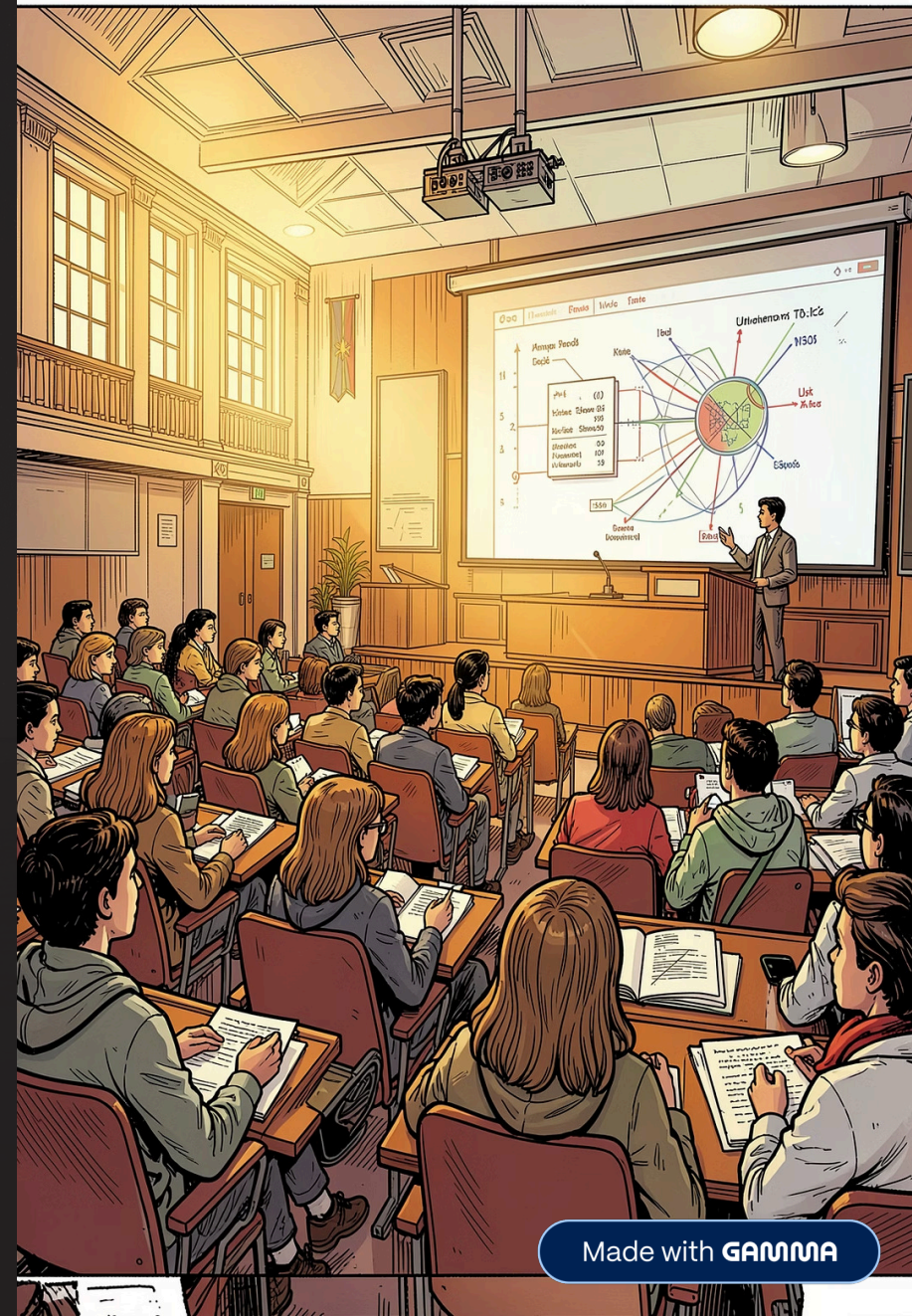
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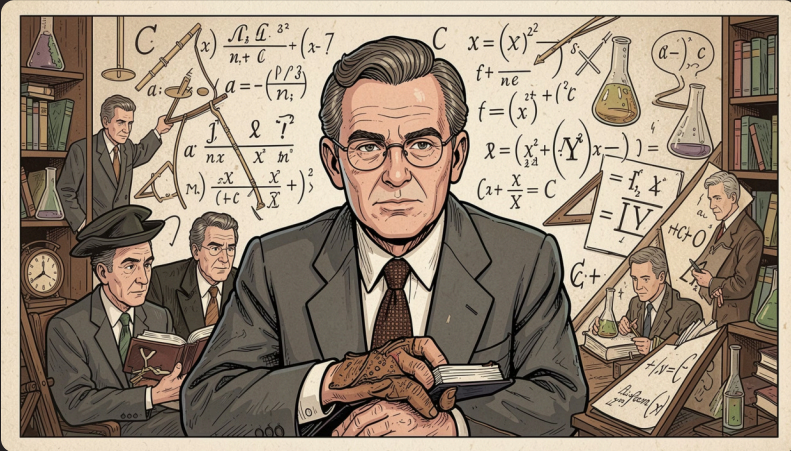
Purpose

This brief presentation compares :

- Hamming Code and Polar Code across working principles
- mathematical formulation
- error correction capability
- complexity, applications, and performance.



Hamming Code – Essentials



What it is

Developed by Richard Hamming (1950). A linear block code designed for single-bit error correction and double-bit error detection. Commonly used in memory systems and low-noise links.

Structure & parity

Notation: (n, k) where k = data bits, n = total bits. Example: $(7, 4)$ has 4 data + 3 parity bits. Parity bits computed with modulo-2 (XOR).

Parity condition

$2^r \geq k + r + 1$ – determines minimum parity bits r .

Parity equations (example)

$$P1 = D1 \oplus D2 \oplus D4$$

$$P2 = D1 \oplus D3 \oplus D4$$

$$P3 = D2 \oplus D3 \oplus D4$$

Polar Code – Essentials



- **The Problem:** Normal channels suffer from moderate noise, slowing down data transfer.
- **The Breakthrough (2009):** Erdal Arıkan invented a mathematical trick to pair and combine these channels.
- **Channel Polarization:** This process forces a sharp split:
 - Some channels become **perfectly clear** (highly reliable).
 - The rest become **complete static** (highly unreliable).

How it Works:

- Transmit the actual data through the **clear channels**.
- Freeze the **noisy channels** by filling them with known zeros.

Encoding

$G_N = B_N \cdot F^{\otimes n}$,
with $F = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$
and $F^{\otimes n}$ the Kronecker power.
Select information bits for reliable channels; freeze others.

Complexity & decoding

Encoding complexity:
 $O(N \log N)$.
Decoding: Successive Cancellation (SC) and Successive Cancellation List (SCL) – accuracy vs. complexity tradeoff.

Applications

Adopted in 5G NR for control channels and other modern wireless standards where strong performance near capacity is required.



Side-by-Side Comparison

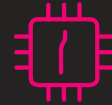
Quick reference table summarizing key parameters for selection in system design.



Type

Hamming: Linear block code ·

Polar: Capacity-achieving modern code



Error correction

Hamming: single-bit correction, double-bit detection ·

Polar: Strong multi-bit correction with suitable decoding



Complexity & latency

Hamming: Low complexity, very low latency ·

Polar: Moderate-high complexity, higher latency with list decoding



Applications

Hamming: memory, simple links ·

Polar: 5G control channels, capacity-sensitive systems

Advantages & Limitations

Hamming Code

• Advantages

- Simple implementation and decoding
- Low hardware cost and low latency
- Good for teaching and memory protection

• Disadvantages

- Only single-bit correction
- Poor performance in high noise

Polar Code

• Advantages

- Near-Shannon performance for large N
- Good performance at low SNR with list decoding
- Standardized for modern wireless systems

• Disadvantages

- Higher decoding complexity and implementation cost
- Increased latency with advanced decoders (SCL)

Practical Guidance for Engineers

01

Choose Hamming when:

System constraints demand minimal complexity, ultra-low latency, or protection of small memory words where only simple single-bit errors are expected.

03

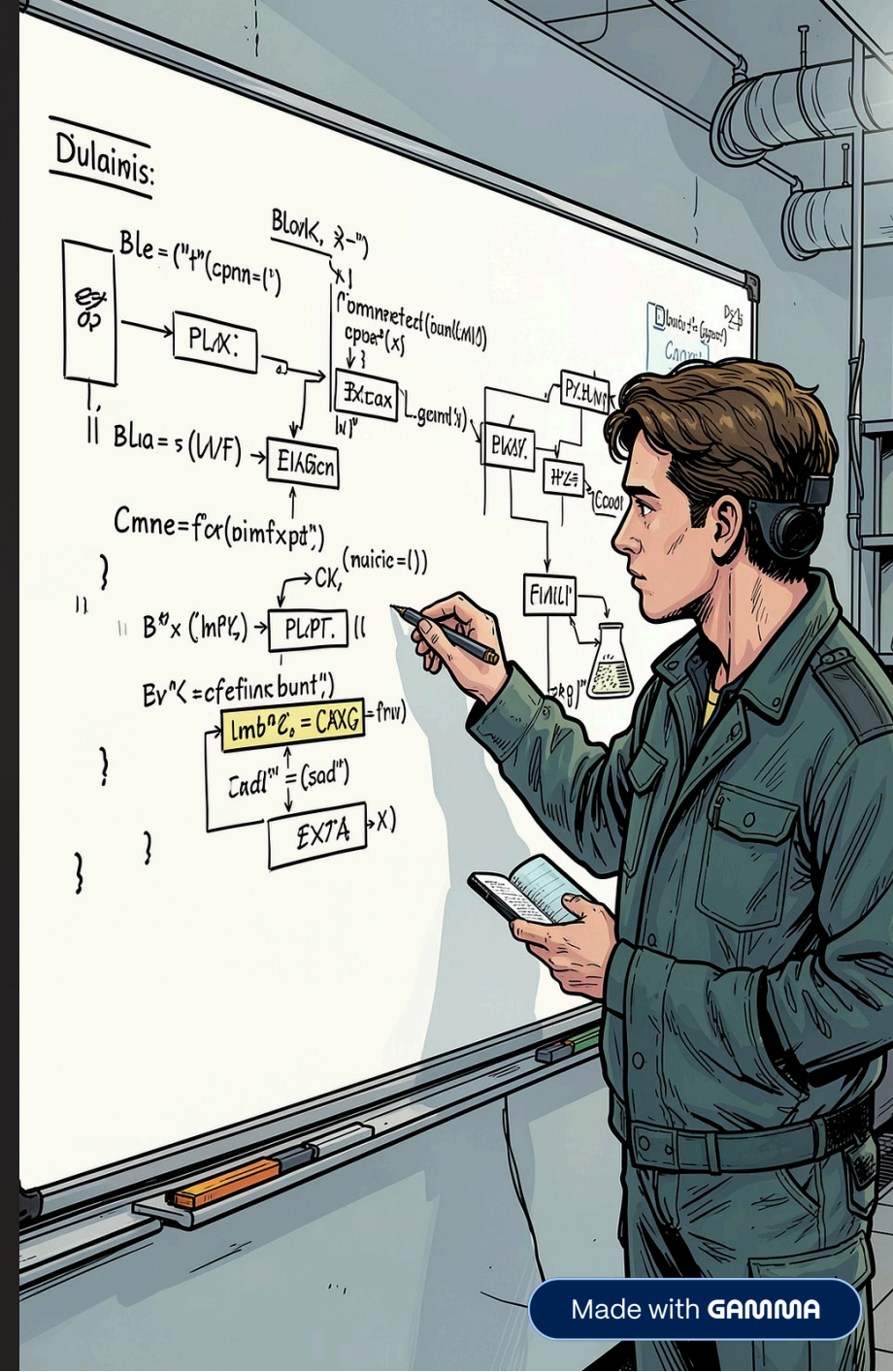
Design tips

Evaluate block length N , decoder choice (SC vs SCL), and hardware resources. Simulate BER vs. SNR for target link-budget before committing to a code.

02

Choose Polar when:

Applications require near-capacity performance, operation at low SNR, or integration with standards (e.g., 5G NR) where reliability outweighs complexity.

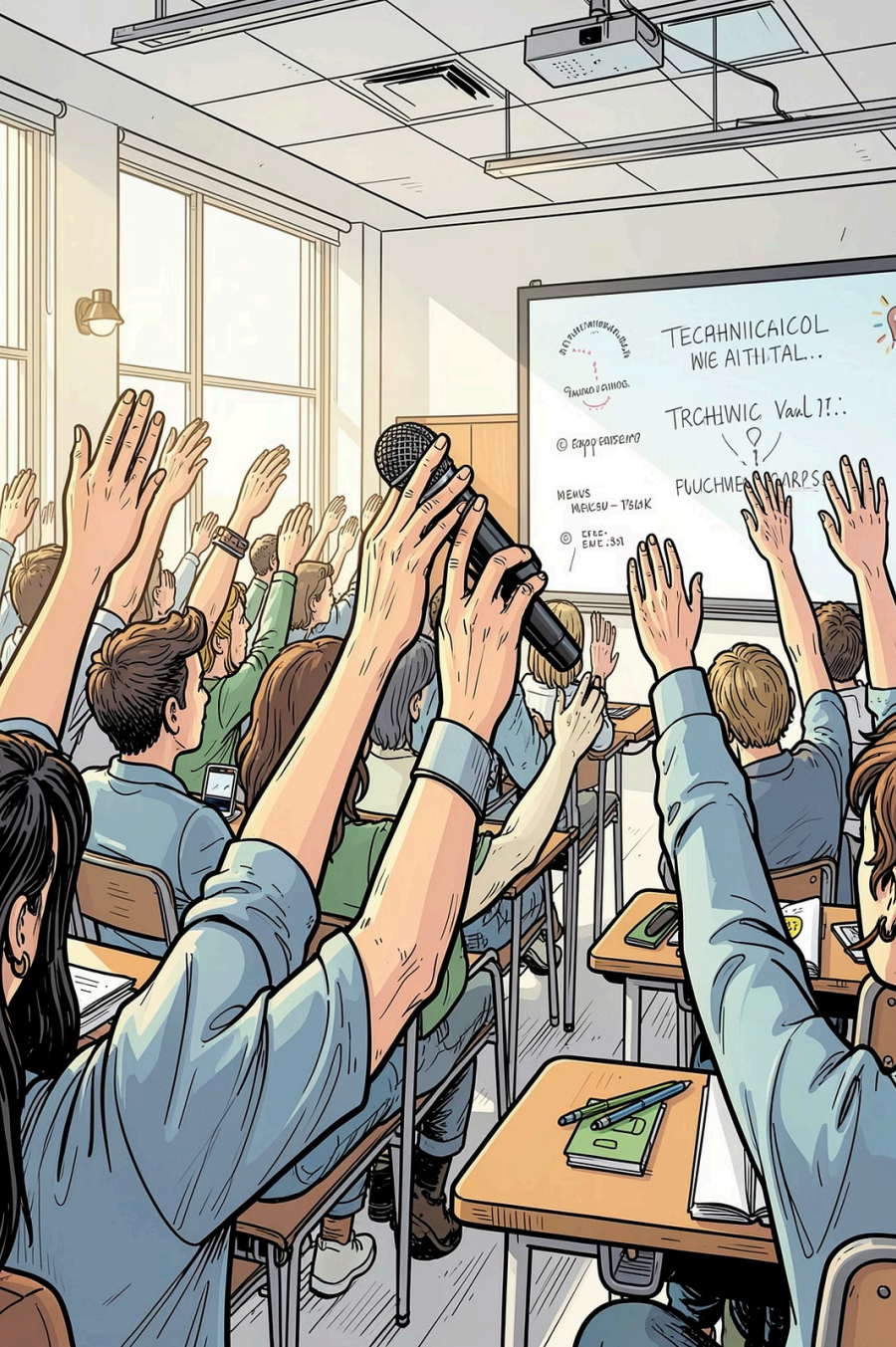


Real-World Applications

Understanding where each code is typically deployed helps in selecting the optimal error correction strategy for specific use cases.

Hamming Code	Polar Code
• Memory Error Correction: Used in computer RAM (e.g., ECC memory) to detect and correct single-bit errors. • Data Storage: Implemented in some hard drives and solid-state drives for data integrity. • Simple Communication Links: Where low-noise environments and minimal complexity are priorities, like early digital communication systems. • Microcontrollers & Embedded Systems: For protecting critical data in constrained environments.	• 5G New Radio (NR): Standardized for control channels in 5G mobile communications, achieving high reliability and spectral efficiency. • Future Wireless Communications: Being explored for various other capacity-sensitive systems where performance close to the Shannon limit is crucial. • High-Throughput Satellite Communication: For robust data transmission in challenging conditions. • Long-Range IoT Devices: Where power efficiency and reliable communication over noisy channels are critical.

While Hamming codes remain vital for their simplicity in specific niches, Polar codes represent the cutting edge for achieving near-optimal performance in modern, high-demand communication systems.



Conclusion & Q&A

Summary:

Hamming = simple, low-cost, low-latency;

Polar = modern, capacity-approaching, better BER at the cost of complexity.

Choose by weighing performance targets, latency, and implementation resources.

 Lets see a simulation

Thank you – ready for questions.